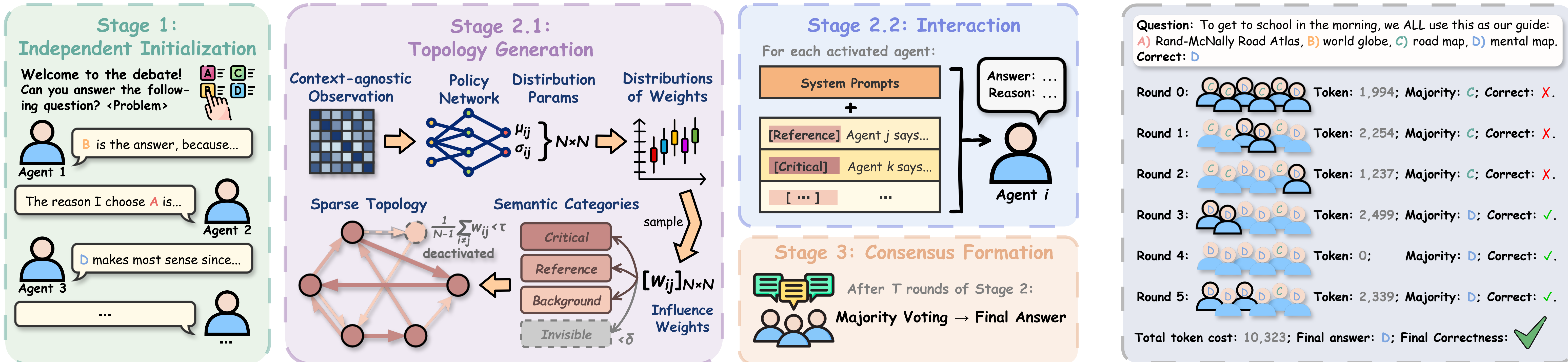


ABSTRACT

Multi-agent debate (MAD) systems leverage collective intelligence to enhance reasoning, yet existing approaches struggle to simultaneously optimize **accuracy**, **consensus formation**, and **computational efficiency**. Static topology methods lack adaptability to task complexity, while external LLM-based coordination risks introducing privileged knowledge that compromises debate neutrality. We present **RUMAD (Reinforcement-Unifying Multi-Agent Debate)**, which formulates dynamic communication topology control as a **reinforcement learning** problem. RUMAD employs a *content-agnostic* observation scheme that captures high-level debate dynamics while avoiding access to raw agent reasoning content, and uses a *multi-objective reward* to model solution quality, cohesion and efficiency. A **PPO-trained controller** dynamically adjusts edge weights in the communication graph, while a **dual-threshold mechanism** enables fine-grained control over both agent activation and information visibility. Across MMLU, GSM8K, and GPQA benchmarks, RUMAD achieves substantial efficiency gains—**>80% token cost reduction**—while **improving reasoning accuracy** over single-LLM and multiple MAD baselines. Notably, RUMAD trained exclusively on MMLU exhibits **robust zero-shot generalization** to out-of-domain (OOD) tasks, indicating that the learned communication strategies capture task-independent principles of effective multi-agent coordination.

PIPELINE & EXAMPLE



(a) **RUMAD Pipeline.** Stage 1 – **Independent Init.** agents independently generate initial responses, preserving diverse hypotheses. Stage 2 – **Adaptive Sparse Debate.** a PPO policy observes debate dynamics and outputs edge weights w_{ij}^t , masking low-influence agents (**node-wise**) and grouping visible neighbors into prompt tiers (**edge-wise**). Stage 3 – **Consensus** via majority voting. (b) **Qualitative Example.** An example debate trace controlled by RUMAD: agents whose own-weight exceeds their average external influence are deemed *confident* and silenced.

MOTIVATION

Three challenges of existing MAD

Adaptivity – static ring/star/group topologies (Sparse MAD, Group Debate, S²-MAD) apply the same budget to every problem, cutting off information paths on hard tasks while retaining redundant edges on easy ones.

Privacy – adaptive methods using external LLM “judgers”/“summarizers” expose debate context to proprietary models; injected external knowledge suppresses internal agent diversity.

Efficiency – most MAD designs optimize only accuracy or consensus, lacking an explicit model of token cost. Static topologies cap communication via empirical rules; external-LLM coordinators yield only an uncontrollable open-loop budget.

Comparison with existing MAD

Design Target	Static Topology	External LLM	Ours
Accuracy	✓	✓	✓
Consensus	●	✓	✓
External-opinion indep.	✓	✗	✓
Question independent	✓	✗	✓
Cost controllable	●	●	✓

✓ targeted ● side-effect ✗ fails

METHOD

A centralized PPO controller regulates inter-agent communication round by round.

Observation – content-agnostic

The controller never accesses raw agent text or embeddings—only a scalar $N \times N$ similarity matrix mixing reasoning-embedding cosine and answer-agreement indicator is exposed:

$$\text{sim}(i, j) = \lambda \cos(r_i^t, r_j^t) + (1 - \lambda) \mathbb{I}(c_i^t = c_j^t)$$

$$o_t = \text{MLP}([\text{sim}(i, j)]_{N \times N})$$

Action – stochastic continuous edges

The policy outputs per-edge $(\mu_{ij}^t, \sigma_{ij}^t)$; weights are sampled from a Sigmoid-squashed Gaussian, yielding a fully differentiable action space.

$$w_{ij}^t \sim \text{Sigmoid}(\mathcal{N}(\mu_{ij}^t, \sigma_{ij}^t)) \in (0, 1)$$

Dual-threshold efficiency

Node-wise: agent i is silenced (reuses previous answer, zero tokens) when $\bar{w}_i^t < w_{ii}^t$ —an implicit self-confidence check. **Edge-wise:** for active agents, j is quantized into four tiers {Critical, Reference, Background, Invisible} from w_{ij}^t .

Budget as training-time prior

A soft penalty on the number of active edges internalizes sparsity in the policy rather than imposing a hard inference-time cap:

$$\mathcal{L}_{\text{budget}}(\theta) = \max(0, \sum_{i,j} \mathbb{I}(w_{ij}^t > \delta) - B)$$

Multi-objective hierarchical reward

$$R_t = \alpha_1 \text{Acc}_t + \alpha_2 \text{Cons}_t + \alpha_3 \text{Prog}_t + \alpha_4 \text{Eff}_t + \alpha_5 \text{Imp}_t - \alpha_6 \text{Spars}_t$$

$$R_{ep} = \beta_1 \text{Acc}_{ep} + \beta_2 \text{Cons}_{ep} + \beta_3 \text{Eff}_{ep} + \beta_4 \text{Imp}_{ep}$$

Step-wise R_t drives runtime adjustment; terminal R_{ep} captures episode-level goals. Trained with PPO + GAE: $\mathcal{L} = \mathcal{L}_\pi + c_1 \mathcal{L}_v + c_2 \mathcal{L}_{\text{budget}}$.

MAIN RESULTS

Task	Method	Accuracy	Tokens (k/task)	Cost Saving
MMLU	RUMAD _{B=12}	68%	11.4	−81.7%
	RUMAD _{B=18}	68%	22.5	−64.1%
	MAD (full)	49%	62.6	N/A
	S-MAD _*	61%	33.3	−46.8%
	S-MAD _o	54%	31.7	−49.3%
	GD	53%	32.2	−48.6%
	S ² MAD	46%	25.4	−58.5%
GPQA	RUMAD _{B=12}	32%	19.5	−66.2%
	RUMAD _{B=18}	35%	33.4	−42.2%
	MAD (full)	30%	57.8	N/A
	S-MAD _*	30%	36.4	−37.1%
	S-MAD _o	34%	38.6	−33.2%
	GD	34%	26.1	−54.9%
	S ² MAD	28%	23.3	−59.8%
GSM8K	RUMAD _{B=12}	86%	10.5	−86.4%
	RUMAD _{B=18}	89%	17.3	−77.5%
	MAD (full)	88%	76.9	N/A
	S-MAD _*	83%	39.4	−48.8%
	S-MAD _o	70%	37.8	−50.8%
	GD	84%	29.8	−61.3%
	S ² MAD	70%	30.5	−60.3%

6 agents, 6 rounds. Trained *only* on MMLU-dev; evaluated zero-shot on GPQA & GSM8K. Agent pool: LLaMA-3.1-8B, ChatGLM-4-9B, DeepSeek-Math-7B (2 each). All results under identical **0-shot, 4-bit** setting.

ABLATION STUDY

